

Lecture 3: From Natural Deduction to Proofs in Lean

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1 Lecture 3: From Natural Deduction to Proofs in Lean

1.1 Aim of the lecture

In the previous lecture, we learned to read a proof state and used tactics such as `intro`, `apply`, `have`, `exact`, and `cases`. We now connect those commands to the rules of natural deduction. Tactics are not arbitrary moves: they build and use proofs according to the logical structure of propositions.

The lecture has five aims:

- revisit the idea of type inhabitation;
- introduce the Curry-Howard correspondence;
- read introduction and elimination rules in natural deduction;
- recognize those rules in Lean proofs;
- distinguish constructive reasoning from classical *reductio ad absurdum*.

1.2 Types and inhabitants

A type describes a class of possible objects. A type is *inhabited* when there is at least one term of that type. If `t` has type `T`, we write `t : T` and say that `t` inhabits `T`.

The idea is not limited to Lean. The informal type “moon of Jupiter” is inhabited because `Io` is a moon of Jupiter. `Io` is a concrete object satisfying the description expressed by the type (Source: [NASA Science, Io: Facts](#)).

```
#check 4           -- 4 : Nat
#check True        -- True : Prop
#check True.intro  -- True.intro : True
```

`Nat` is inhabited, for example, by `4`. `True` is also inhabited: `True.intro` has exactly type `True`. This observation leads from the general relation between terms and types to the relation between proofs and propositions.

1.3 The Curry-Howard correspondence

Under the Curry-Howard correspondence, propositions and types are two ways of describing the same structure. A proposition specifies what must be proved; the corresponding type

specifies what kind of term must be built. A proof of the proposition is a term inhabiting that type (Source: [Wadler, Propositions as Types](#)).

Logic	Type theory
proposition	type
proof	term
$A \rightarrow B$	function from proofs of A to proofs of B
introduction of $\rightarrow$	function construction
elimination of $\rightarrow$	function application

Checking a proof therefore means checking that the constructed term has the declared type. A derivation does not merely certify that a proposition is true: it specifies how to construct its proof (Source: [Theorem Proving in Lean 4, Propositions and Proofs](#)).

1.4 Natural deduction and the proof state

Natural deduction presents a proof as a sequence of local steps governed by rules. Each rule shows how to obtain a conclusion from premises. Introduction rules explain how to construct a proof with a particular main connective; elimination rules explain how to use an available proof containing that connective (Source: [Logic and Proof, Natural Deduction for Propositional Logic](#)).

A judgment has the form $\Gamma \vdash A$. Here Γ is the context of currently available assumptions and A is the conclusion to derive. Lean’s proof state has the same organization: the context appears above the turnstile and the goal after it.

Some rules introduce temporary assumptions. To prove $A \rightarrow B$, for example, we temporarily assume A and construct B . Once the implication is complete, the assumption has been incorporated into a function from proofs of A to proofs of B .

1.5 Implication

1.5.1 Implication introduction, ‘ $\rightarrow I$ ’

$$\begin{array}{c}
 \frac{}{A} \quad 1 \\
 \vdots \\
 \frac{B}{A \rightarrow B} \quad 1 \rightarrow \\
 \text{Implication introduction, } \rightarrow I
 \end{array}$$

To prove $A \rightarrow B$, we introduce an assumption of type A and construct a proof of B . In Lean, `intro` performs precisely this step.

We want to prove that if it rains and it is cold, then it rains. We introduce the assumption that it rains and is cold; we obtain that it rains by taking the left component; we use exactly that fact.

```

theorem lecture03_imp_intro
  (Rains IsCold : Prop) :
    Rains ∧ IsCold → Rains := by
  intro hRainsAndIsCold
  have hRains := hRainsAndIsCold.left
  exact hRains

```

1.5.2 Implication elimination, '→E'

$$\frac{A \rightarrow B \quad A}{B} \rightarrow E$$

Implication elimination, →E

If we have $A \rightarrow B$ and A , we apply the first proof to the second and obtain B . This is modus ponens.

```

theorem lecture03_imp_elim
  (Rains TakeUmbrella : Prop)
  (hRainsUmbrella : Rains → TakeUmbrella)
  (hRains : Rains) :
    TakeUmbrella := by
  have hTakeUmbrella := hRainsUmbrella hRains
  exact hTakeUmbrella

```

`hRainsUmbrella` behaves as a function: it receives a term of type `Rains` and returns a term of type `TakeUmbrella`.

1.6 Conjunction

1.6.1 Conjunction introduction, '∧I'

$$\frac{A \quad B}{A \wedge B} \wedge I$$

Conjunction introduction, ∧I

To prove $A \wedge B$, we must construct both components. Applying `And.intro` splits the goal into two cases: first A , then B .

```

theorem lecture03_and_intro
  (Rains IsCold : Prop)
  (hRains : Rains)
  (hIsCold : IsCold) :
  Rains ∧ IsCold := by
  apply And.intro
  · exact hRains
  · exact hIsCold

```

The bullets belong to distinct goals, corresponding to the two conjuncts.

1.6.2 Conjunction elimination, ‘ $\wedge E_l$ ’ and ‘ $\wedge E_r$ ’

$$\frac{A \wedge B}{A} \wedge E_l$$

Left conjunction elimination, $\wedge E_l$

$$\frac{A \wedge B}{B} \wedge E_r$$

Right conjunction elimination, $\wedge E_r$

A proof of $A \wedge B$ contains a proof of A and a proof of B . We select the left component with `.left` and the right component with `.right`.

```

theorem lecture03_and_elim_left
  (Rains IsCold : Prop) (hBoth : Rains ∧ IsCold) : Rains := by
  have hRains := hBoth.left
  exact hRains

```

```

theorem lecture03_and_elim_right
  (Rains IsCold : Prop) (hBoth : Rains ∧ IsCold) : IsCold := by
  have hIsCold := hBoth.right
  exact hIsCold

```

Introduction constructs a conjunction from its components; elimination travels in the opposite direction and retrieves a component.

1.7 Disjunction

1.7.1 Disjunction introduction, 'vI_l' and 'vI_r'

$$\frac{A}{A \vee B} \vee I_l$$

Left disjunction introduction, vI_l

$$\frac{B}{A \vee B} \vee I_r$$

Right disjunction introduction, vI_r

To prove $A \vee B$, it is enough to construct one side, but we must specify which one. `Or.inl` chooses the left side; `Or.inr` chooses the right side.

```
theorem lecture03_or_intro_left
  (Rains Snows : Prop) (hRains : Rains) : Rains v Snows := by
  apply Or.inl
  exact hRains
```

```
theorem lecture03_or_intro_right
  (Rains Snows : Prop) (hSnows : Snows) : Rains v Snows := by
  apply Or.inr
  exact hSnows
```

1.7.2 Disjunction elimination, 'vE'

$$\begin{array}{c}
 \frac{\frac{\overline{A}^1 \quad \overline{B}^1}{\vdots} \quad \frac{\overline{B}^1 \quad \overline{A}^1}{\vdots}}{A \vee B} \quad \frac{C \quad C}{C} \quad 1 \vee E \\
 \hline
 C
 \end{array}$$

Disjunction elimination, vE

If we have $A \vee B$, we do not know which side was proved. To obtain one conclusion C , we must derive it both in the A case and in the B case.

```

theorem lecture03_or_elim
  (Rains Snows TakeUmbrella : Prop)
  (hRainsOrSnows : Rains v Snows)
  (hRainsUmbrella : Rains → TakeUmbrella)
  (hSnowsUmbrella : Snows → TakeUmbrella) :
  TakeUmbrella := by
cases hRainsOrSnows with
| inl hRains =>
  have hTakeUmbrella := hRainsUmbrella hRains
  exact hTakeUmbrella
| inr hSnows =>
  have hTakeUmbrella := hSnowsUmbrella hSnows
  exact hTakeUmbrella

```

`cases` eliminates the disjunction by making all possible forms explicit.

1.8 Negation, truth, and falsity

1.8.1 Negation

In Lean, $\neg A$ is defined as $A \rightarrow \text{False}$. A proof of $\neg A$ is therefore a function taking every hypothetical proof of A to a contradiction.

$$\begin{array}{c}
 \overline{A}^1 \\
 \vdots \\
 \frac{\perp}{\neg A} \quad 1 \neg I \\
 \hline
 \neg A
 \end{array}$$

Negation introduction, $\neg I$

To introduce $\neg A$, we introduce A and construct `False`.

```

theorem lecture03_not_intro
  (DrinkCoffee StayAwake : Prop)
  (hCoffeeAwake : DrinkCoffee → StayAwake)
  (hNotAwake : ¬StayAwake) : ¬DrinkCoffee := by
intro hDrinkCoffee
have hStayAwake := hCoffeeAwake hDrinkCoffee
have hContradiction : False := hNotAwake hStayAwake
exact hContradiction

```

$$\frac{\neg A \quad A}{\perp} \neg E$$

Negation elimination, $\neg E$

To eliminate a negation, we apply $\neg A$ to A and obtain False.

```

theorem lecture03_not_elim
  (LabOpen : Prop) (hOpen : LabOpen) (hNotOpen : ¬LabOpen) : False := by
have hContradiction := hNotOpen hOpen
exact hContradiction

```

1.8.2 Truth and falsity

$$\frac{}{\top} \top I$$

Truth introduction, $\top I$

True has a canonical constructor and requires no assumptions.

```

theorem lecture03_true_intro : True := by
exact True.intro

```

`exact True.intro` can also be replaced by `trivial`.

$$\frac{\perp}{A} \perp E$$

Falsity elimination, $\perp E$

False has no constructors. If the context already contains a proof of False, we can eliminate falsity and obtain any proposition. This is the principle of *ex falso quodlibet*.

```

theorem lecture03_false_elim
  (Rains : Prop) (hContradiction : False) : Rains := by
  apply False.elim
  exact hContradiction

```

The rule does not let us prove anything arbitrarily: it applies only after a contradiction has been constructed.

1.9 Biconditional

A proof of $A \leftrightarrow B$ contains two functions: one from A to B and one from B to A .

1.9.1 Biconditional introduction, ' $\leftrightarrow I$ '

$$\begin{array}{ccc}
 \overline{A} & 1 & \overline{B} & 1 \\
 & \vdots & & \vdots \\
 B & & A & \\
 \hline
 A \leftrightarrow B & 1 & \leftrightarrow &
 \end{array}$$

Biconditional introduction, $\leftrightarrow I$

To prove that “it rains and it is cold” is equivalent to “it is cold and it rains,” we construct both directions.

```

theorem lecture03_iff_intro
  (Rains IsCold : Prop) :
  Rains ∧ IsCold ↔ IsCold ∧ Rains := by
  apply Iff.intro
  · intro hRainsAndIsCold
    apply And.intro
    · exact hRainsAndIsCold.right
    · exact hRainsAndIsCold.left
  · intro hIsColdAndRains
    apply And.intro
    · exact hIsColdAndRains.right
    · exact hIsColdAndRains.left

```


1.9.2 Biconditional elimination, ‘ $\leftrightarrow E_l$ ’ and ‘ $\leftrightarrow E_r$ ’

$$\frac{A \leftrightarrow B \quad A}{B} \leftrightarrow E_l$$

Left-to-right biconditional elimination, $\leftrightarrow E_l$

$$\frac{A \leftrightarrow B \quad B}{A} \leftrightarrow E_r$$

Right-to-left biconditional elimination, $\leftrightarrow E_r$

`Iff.mp` extracts the left-to-right direction; `Iff.mpr` extracts the right-to-left direction.

```
theorem lecture03_iff_elim_left
  (EvenNumber DivisibleByTwo : Prop)
  (hEquivalence : EvenNumber  $\leftrightarrow$  DivisibleByTwo)
  (hEvenNumber : EvenNumber) : DivisibleByTwo := by
  have hDirection := Iff.mp hEquivalence
  have hDivisibleByTwo := hDirection hEvenNumber
  exact hDivisibleByTwo
```

```
theorem lecture03_iff_elim_right
  (EvenNumber DivisibleByTwo : Prop)
  (hEquivalence : EvenNumber  $\leftrightarrow$  DivisibleByTwo)
  (hDivisibleByTwo : DivisibleByTwo) : EvenNumber := by
  have hDirection := Iff.mpr hEquivalence
  have hEvenNumber := hDirection hDivisibleByTwo
  exact hEvenNumber
```

After extracting the needed direction, we apply it as an ordinary implication.

1.10 Constructive rules and reductio ad absurdum

The rules considered so far have a direct constructive reading: every proof specifies how to build the required term. In Lean’s constructive base logic, excluded middle $A \vee \neg A$ and double-negation elimination $\neg\neg A \rightarrow A$ are not automatically available. Lean allows classical principles to be used locally and explicitly (Source: [Theorem Proving in Lean 4, Classical Logic](#)).

Classical reductio ad absurdum proves A by showing that assuming $\neg A$ leads to `False`. In Lean, we apply it with `Classical.byContradiction`.

$$\frac{\frac{\vdots}{\perp} \quad 1}{A} \text{ RAA}$$

Reductio ad absurdum, RAA

```
theorem lecture03_reductio_ad_absurdum
  (Rains TakeUmbrella : Prop)
  (hNotRainsNotUmbrella : ¬Rains → ¬TakeUmbrella)
  (hTakeUmbrella : TakeUmbrella) : Rains := by
  apply Classical.byContradiction
  intro hNotRains
  have hNotTakeUmbrella := hNotRainsNotUmbrella hNotRains
  have hContradiction := hNotTakeUmbrella hTakeUmbrella
  exact hContradiction
```

The first assumption turns $\neg \text{Rains}$ into $\neg \text{TakeUmbrella}$. The latter is a function from `TakeUmbrella` to `False`; applying it to `hTakeUmbrella` produces the required contradiction.

1.11 Choosing a rule

When reading a proof state, ask two questions. First: what is the main connective of the goal? It often suggests an introduction rule. Second: which compound proofs are already in the context? They suggest elimination rules.

Lean tactic or term	Natural-deduction reading
<code>intro h</code>	introduce an assumption
<code>apply C</code>	apply a constructor or rule
<code>have h := ...</code>	obtain an intermediate fact
<code>exact h</code>	use exactly the required term
<code>cases h with</code>	consider the possible cases
<code>h.left, h.right</code>	eliminate a conjunction
<code>hAB hA</code>	eliminate an implication

This vocabulary describes the structure of the term Lean is constructing and makes the relationship between formal rule, natural-language argument, and proof script explicit.

1.12 Working strategy

When constructing a proof:

1. read the goal and assumptions; 2. identify the goal's main connective; 3. choose an introduction rule when the goal suggests one; 4. look for a useful elimination rule in the context; 5. name important intermediate results with `have`; 6. close the goal with `exact` when the required term is available; 7. verify that every case and subgoal has been solved.

`Classroom.lean` presents each rule separately. `Exercises.lean` begins with direct applications and then combines rules into complete arguments. `Solutions.lean` keeps the proof structure explicit so that each Lean step can be connected to its natural-deduction rule.

1.13 Sources and further reading

- Jeremy Avigad, Leonardo de Moura, Soonho Kong, and Sebastian Ullrich, *Theorem Proving in Lean 4*, “Propositions and Proofs”: https://lean-lang.org/theorem_proving_in_lean4/Propositions-and-Proofs/.
- Jeremy Avigad, Robert Y. Lewis, and Floris van Doorn, *Logic and Proof*, “Natural Deduction Rules”: https://leanprover.github.io/logic_and_proof_lean3/nd_quickref.html.
- Philip Wadler, “Propositions as Types,” *Communications of the ACM* 58(12), 2015: <https://homepages.inf.ed.ac.uk/wadler/papers/propositions-as-types/stanford.pdf>.
- NASA Science, “Io: Facts”: <https://science.nasa.gov/jupiter/jupiter-moons/io/facts/>.